

Bill of Materials

Item	Component	Arduino Connection Pins	Communication Interface	Operating Voltage	Function	Qty.	Approx. Cost (USD)	Product Reference
1	Geekcreit Arduino Uno R3-Compatible Microcontroller (ATmega328P)	USB Type-B	USB, I ² C, SPI, UART	5 V	Central processing unit responsible for sensor interfacing, data acquisition, processing and data logging.	1	10.0	https://www.banggood.com/GEEKCREIT-UNO-R3-ATmega328P-Development-Board-Compatible-with-Arduino-p-964163.html
2	Sensirion SCD30 CO ₂ , Temperature and Relative Humidity Sensor	VCC→5 V; GND→GND; SDA→SDA; SCL→SCL	I ² C	5.5 V	Measures atmospheric CO ₂ concentration, air temperature and relative humidity.	1	41.13	https://sensirion.com/products/catalog/SCD30
3	Adafruit SHT40 Temperature and Relative Humidity Sensor	VCC→5 V; GND→GND; SDA→A4 (SDA); SCL→A5 (SCL)	I ² C	5 V*	Measures air temperature and relative humidity.	1	5.95	https://www.adafruit.com/search?q=sht40
4	DFRobot Fermion microSD Card Module (DFR0229)	VCC→5 V; GND→GND; CS→D10; MOSI→D11; MISO→D12; SCK→D13	SPI	3.3-5 V	Stores environmental measurements on a microSD card in CSV format.	1	5.20	https://www.dfrobot.com/product-875.html
5	SanDisk Ultra microSD Card (16 GB, Class 10)	Inserted into microSD module	SPI (via module)	—	Stores recorded environmental measurements.	1	8.00	https://www.sandisk.com/products/memory-cards/microsd-cards/sandisk-ultra-uhs-i-microsd
6	Jumper Wires	Power and signal connections	Electrical	—	Provides electrical connections between components.	1 set	4.00	https://www.sparkfun.com/products/11026
7	USB Type-A to Type-B Cable	USB Type-B	USB 2.0	5 V	Supplies power and enables programming and serial communication.	1	3.00	https://www.startech.com/en-us/cables/usb2hab1m
8	Computer (Windows 11)	USB	USB	—	Programming, serial monitoring and data retrieval.	1	—	Existing laboratory equipment
Total Approximate cost							77.28	